Network Performance Analysis — Tutorial 3
Erlang Delay and Loss and $M/G/1$ Queue

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1. A telephone call centre services complaints from bank customers about the quality of their service. They expect at 'busy hour' calls to arrive at a rate of 120 customers an hour. Dealing with each customer takes on average 3 minutes. When all serving assistants are occupied customers are placed in a holding system.

   (a) The service center has 10 serving assistants. What is the probability of a customer having to join the queue?

   (b) What is the expected queueing time for each customer?

   (c) Given that customers who don't have to queue have an average queueing time of 0, what is the average waiting time for those who do have to queue?

   (d) The bank wishes to combine two of these service centers for efficiency. Both service centers have arrival rates of 120 customers an hour in 'busy hour'. How many staff will they need at the new centre to maintain a queueing probability at least as low as in a)?

   (e) Answer parts b) and c) for the new system.

   For this question you may assume that arrivals are Poisson distributed and service times are exponentially distributed.

2. A telephone company plans to rent trunk lines between Birmingham and London for \$1,000 per year. They expect to have a demand of 20 calls per hour, and an average holding time of 15 minutes. They can charge 0.01\$/minute for each customer. They will provide no holding facility.

   (a) How many trunk lines will they need to keep the blocking probability below 0.01?

   (b) How much profit will they make?

   For this question you may assume that arrivals are Poisson distributed and service times are exponentially distributed.

3. A service time distribution is defined as follows:

$$P(X) = \begin{cases} \frac{2}{b} - \frac{2}{b^2}X & 0 < X \leq b \\ 0 & X > b \end{cases} \quad (1)$$

(a) Sketch this distribution.

(b) The Pollaczek-Khinchin formula states the average waiting time for a queue is given by

$$\bar{W} = \frac{\lambda \left\langle X^2 \right\rangle}{2(1-\rho)} \tag{2}$$

(c) By determining the first two moments, $\langle X \rangle$ and $\left\langle X^2 \right\rangle$, of the distribution given in 1 derive the average waiting time for the queue.